

Summary: Financial Markets and Institutions, Chapter 4 (pp. 107-116)

Saunders, Cornett & Erhemjamts, 8th ed. - The Federal Reserve System, Monetary Policy, and Interest Rates

Core message

The Fed influences the money supply, interest rates and the wider economy mainly by changing the **supply of or demand for bank reserves**. A change in excess reserves feeds through to short- and long-term rates, exchange rates, credit, employment, output and prices.

- **Fed funds rate:** the overnight interbank lending rate and the Fed's main policy target. The FOMC has targeted it since 1993.
- **IOER / IOER:** the Fed pays interest on required and excess reserves, giving it a way to keep the fed funds rate inside its target range. In Dec 2015 the target range was 0.25-0.50% and IOER was 0.50%. The target rose nine times through Dec 2018, to 2.25-2.50%; IOER went from 0.50% to 2.40%.
- **Two approaches:** target the quantity of reserves (monetary base) or the price of reserves (fed funds rate). The Fed has mostly used the rate.
- **Leading indicators (Table 4-4):** weekly manufacturing hours, jobless claims, new orders, building permits, S&P 500, M2, yield spread, consumer sentiment.

Rate history

Period	Move
2001	11 rate cuts after the dot-com collapse and 9/11
2000-2003	Target cut from 6.5% to 1.0%
July 2004 - Aug 2006	17 straight 0.25-point hikes, from 1% to 5.25%
Sept 2007 - Dec 16, 2008	From 5.25% to a 0-0.25% range; discount rate 0.50%
Dec 2015	First hike in 10 years
Dec 2018	Target 2.25-2.50% after nine 0.25-point hikes
Aug 1, 2019	Cut to 2.00-2.25%

- **Housing bubble:** the text argues very low rates in 2000-2003 (arguably below true inflation) helped fuel the bubble; the 2004-06 hikes raised ARM payments and contributed to its deflation.
- **Sept 17, 2019 repo spike:** fed funds hit 5% vs a 2.00-2.25% target; repo rates reached about 10% (from just over 2%). Banks were near their minimum reserves. The Fed injected billions into repo for the first time in over a decade and resumed buying Treasuries at about \$60 billion a month. Analysts had put the remaining cushion at only \$200-300 billion.

Tool 1: Open market operations

- The FOMC sends a **policy directive** to the New York Fed's Trading Desk, which trades with primary dealers (24 dealers, about \$1.27 trillion a day as of Dec 2019).
- Fed buys securities: bank reserves rise and lending capacity expands. Fed sells: reserves fall.
- **Examples 4-1 / 4-2:** a \$500 million purchase adds \$500 million to the Fed's Treasury holdings, to bank reserves, and to dealers' deposits at commercial banks. A sale does the reverse.

Program	Details
Crisis response	Balance sheet grew from \$887 billion to \$2.2 trillion in fall 2008

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QE1 (from Mar 2009)	\$300 billion long-term Treasuries over 6 months, plus \$1.25 trillion agency MBS and bonds (Jan 2009 - Mar 2010)
QE2 (Nov 2010 - Jun 2011)	\$600 billion long-term Treasuries
QE3 (Sept 2012 - Oct 2014)	Open-ended, \$40 billion a month

- Treasury holdings: **\$2.46 trillion** in 2016, up 232% from \$741 billion in 2007.
- Depository institution reserves: **\$1.98 trillion**, up 9,406% from \$21 billion in 2007.
- Agency securities: **\$1.78 trillion** in 2016, up from \$0 in 2007.
- Normalization (Oct 2017 - Aug 2019): Treasuries fell to **\$2.34 trillion** in 2018; agency holdings to **\$1.64 trillion**.
- **Why Treasuries:** a highly liquid, deep market that absorbs large trades without big price moves. Before QE, the Fed had not bought long-term Treasuries since the 1960s.
- **Miscellaneous assets:** 4.0% of Fed assets in 2018 but 46.7% in 2008, reflecting crisis programs (AIG loan, commercial paper support).

Tool 2: The discount rate

- The rate the Fed charges depository institutions. Its main role is as a **signal**: raising it signals tightening, lowering it signals easing.
- **Rarely used as a tool because:** (1) the Fed cannot predict how much banks will borrow (little effect in Aug 2007); (2) signaling effects can be huge: the surprise cut to **0.50% on Dec 16, 2008** coincided with a **359.61-point Dow jump**.
- The rate stayed at 0.50% until Feb 2010, when it was raised to 0.75%.
- **January 2003 reform:** primary credit for healthy banks at 1 point *above* the fed funds target (previously below); secondary credit for troubled banks at 1.5 points above; seasonal credit for small agricultural/tourist-area banks. Healthy banks no longer have to prove they cannot borrow elsewhere, reducing stigma.
- **Crisis expansion:** the spread over the fed funds target was cut to 0.25 points. Investment banks gained access via the Primary Dealer Credit Facility in spring 2008, borrowing an average of **\$31.3 billion a day** in the first three days. In Aug 2007, four major banks borrowed **\$2 billion** to reduce the stigma.
- **Figure 4-4:** prime, CD, T-bill and discount rates move together.

Tool 3: Reserve requirements

Only introduced before the excerpt ends: reserve requirements set the minimum reserve assets (vault cash plus bank deposits at the Fed).

Main takeaway

The Fed's tools moved from a small, rate-focused operation before 2008 to large balance-sheet programs during and after the crisis, which it then began to unwind.